Curriculum Structure for B. Tech in Computer Science and Engineering
For Academic Year 2024-2025

Third Year - Fifth Semester

Semester V							
Sl. No.	**Type of course**	**Code**	**Course Title**	**Hours per week**			**Credits**
				L	**T**	**P**	
Theory							
1	Engineering Science Course	ESC-CSE 501	Compiler design	3	0	0	3
2	Professional Core Course	PCC-CSE 501	Operating Systems	3	0	0	3
3	Professional Core Course	PCC-CSE 502	Design & Analysis of Algorithms	3	0	0	3
4	Professional Elective courses	PEC-CSE 501 A/B/C	(Elective-I) Artificial Intelligence/ Human Computer Interaction/ Computer Graphics	3	0	0	3
5	Open Elective courses	OEC-CSE 501 A/B/C	(Open Elective-I) Operations Research/ Multimedia Systems/ Introduction to Philosophical Thoughts	3	0	0	3
6	Humanities & Social Sciences including Management course	HSMC-CSE 501	Project Management and Entrepreneurship (Humanities-IV)	2	1	0	3
Practical							
7	Professional Core Course	PCC-CSE 591	Operating Systems Lab	0	0	4	2
8	Professional Core Course	PCC-CSE 592	Design & Analysis of Algorithms Lab	0	0	4	2
9	Professional Core Course	PCC-CSE 593	IT Workshop-II (Web Programming)	0	0	4	2
Total credit of the Semester							**24**

Detailed Syllabus for B. Tech in Computer Science & Engineering
For Academic Year 2024-2025

CSE
Third Year - Fifth Semester

Compiler Design
Code: ESC-CSE 501
Contact: 3L
Theory: 3 hrs. / Week
Credit Points: 3

Course Objective:	
1.	To understand and list the different stages in the process of compilation.
2.	Identify different methods of lexical analysis
3.	Design top-down and bottom-up parsers
4.	Identify synthesized and inherited attributes
5.	Develop syntax directed translation schemes
6.	Develop algorithms to generate code for a target machine

Pre-Requisite:	
1.	Knowledge of automata theory, context free Languages
2.	Computer organization
3.	Data structures and simple graph algorithms, logic or algebra

Unit	Content	Hrs.
1.	**Introduction to Compiling:** Compilers, Analysis of the source program, The phases of the compiler, Cousins of the compiler. The role of the lexical analyser, Tokens, Patterns, 6 Lexemes, Input buffering, Specifications of a token, Recognition of a tokens, Finite automata, From a regular expression to an NFA, From a regular expression to NFA, From a regular expression to DFA, Design of a lexical analyser generator (Lex).	6
2.	**Syntax Analysis:** The role of a parser, Context free grammars, Writing a grammar, Top down Parsing, Non recursive Predictive parsing (LL), Bottom up parsing, Handles, Viable prefixes, Operator precedence parsing, LR parsers (SLR, LALR), Parser generators (YACC). Error Recovery strategies for different parsing techniques.	10
3.	**Syntax directed translation and Type checking:** Syntax director definitions, Construction of syntax trees, Bottom-up evaluation of S attributed definitions, L attributed definitions, Bottom-up evaluation of inherited attributes. Type systems, Specification of a simple type checker, Equivalence of type expressions, Type conversions.	5
4.	**Run time environments:** Source language issues (Activation trees, Control stack, scope of declaration, Binding of names), Storage organization (Subdivision of run-time memory, Activation records), Storage allocation strategies, Parameter passing (call by value, call by reference, copy restore, call by name), Symbol tables, dynamic storage allocation techniques.	7

Pre-Requisite:	
1.	Computer Organization &Architecture

Unit	Content	Hrs.
1.	**Introduction**: Generations Concept of Operating systems, Systems, Types of 3 Operating Systems, OS Services, System Calls, Structure of an OS, Concept of Virtual Machine. Computer system structures.	3
2.	**Processes**: Definition, Process Relationship, Different states of a Process, Process State transitions, Process Control Block (PCB), Context switching. **Thread**: Definition, Various states, Benefits of threads, Types of threads, Concept of multithreads. **Process Scheduling**: Foundation and Scheduling objectives, Types of Schedulers, Scheduling criteria: CPU utilization, Throughput, Turnaround Time, Waiting Time, Response Time; Scheduling algorithms: Pre-emptive and Non pre-emptive, FCFS, SJF, RR; Multiprocessor scheduling: Real Time scheduling: RM and EDF.	10
3.	**Inter-process Communication**: Critical Section, Race Conditions, Mutual Exclusion, Hardware Solution, Strict Alternation, Peterson's Solution, The Producer Consumer Problem, Semaphores, Event Counters, Monitors, Message Passing, Classical IPC Problems: Reader's & Writer Problem, Dinning Philosopher Problematic.	5
4.	**Deadlocks**: Definition, Necessary and sufficient conditions for Deadlock, Deadlock Prevention, Deadlock Avoidance: Banker's algorithm, Deadlock detection and Recovery.	5
5.	**Memory Management**: Basic concept, Logical and Physical address map, Memory allocation: Contiguous Memory allocation– Fixed and variable partition– Internal and External fragmentation and Compaction; Paging: Principle of operation –Page allocation Hardware support for paging, Protection and sharing, Disadvantages of paging. **Virtual Memory**: Basics of Virtual Memory – Hardware and control structures – Locality of reference, Page fault , Working Set , Dirty page/Dirty bit – Demand paging, Page Replacement algorithms: Optimal, First in First Out (FIFO), Second Chance (SC), Not recently used (NRU) and Least Recently used(LRU).	8
6.	**File Management**: Concept of File, Access methods, Allocation methods (contiguous, linked, indexed), Free-space management (bit vector, linked list, grouping). **Disk Management**: Disk structure, Disk scheduling - FCFS, SSTF, SCAN, C-SCAN, Disk reliability, Disk formatting, Boot-block, Bad blocks	6

Text book and Reference books:

1. Operating System Concepts Essentials, 9th Edition by Avi Silberschatz, Peter Galvin, Greg Gagne, Wiley Asia Student Edition.
2. Operating Systems: Internals and Design Principles, 5th Edition, William Stallings, Prentice Hall of India.
3. Operating System Concepts, Ekta Walia, Khanna Publishing House (AICTE Recommended Textbook – 2018)
4. Operating System: A Design-oriented Approach, 1st Edition by Charles Crowley, Irwin Publishing
5. Operating Systems: A Modern Perspective, 2nd Edition by Gary J. Nutt, Addison Wesley
6. Design of the Unix Operating Systems, 8th Edition by Maurice Bach, Prentice-Hall of India
7. Understanding the Linux Kernel, 3rd Edition, Daniel P. Bovet, Marco Cesati, O'Reilly and Associates

Course Outcomes:	
On completion of the course students will be able to	
PCC-CSE 501.1	Understand basic concepts of operating system and process management.
PCC-CSE 501.2	Analyse various scheduling algorithms and process synchronization.
PCC-CSE 501.3	Evaluate various deadlock prevention and avoidance algorithms.
PCC-CSE 501.4	Design various memory, file and disk management schemes.

Mapping of COs with POs and PSOs (Course Articulation Matrix):															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	1	2	1	1	1	1	1	1	-	2	2	1	2
CO2	3	3	3	3	2	-	1	-	2	2	2	2	3	3	3
CO3	2	2	2	3	2	-	-	-	2	2	1	2	2	2	3
CO4	2	3	2	3	2	-	1	1	3	2	2	3	2	3	3
AVG.	2.25	2.25	2.00	2.75	1.75	0.25	0.75	0.50	2.00	1.75	1.25	2.25	2.25	2.25	2.75

Design and Analysis of Algorithms
Code: PCC-CSE 502
Contacts: 3L
Theory: 3 hrs. / Week
Credit Points: 3

Course Objective:	
1.	The aim of this module is to learn how to develop efficient algorithms for simple computational tasks and reasoning about the correctness of them.
2.	Through the complexity measures, different range of behaviours of algorithms and the notion of tractable and intractable problems will be understood.

Pre-Requisite:	
1.	To know data-structure and basic programming ability.

Unit	Content	Hrs.
1.	Introduction: Characteristics of algorithm. Analysis of algorithm: Asymptotic analysis of complexity bounds – best, average and worst-case behaviour; Performance measurements of Algorithm, Time and space trade-offs, Analysis of recursive algorithms through recurrence relations:Substitution method, Recursion tree method and Masters' theorem.	8
2.	**Divide and Conquer and Greedy Approach** Basic method, use, Examples – Binary Search, Merge Sort, Quick Sort and their complexity , Strassen's matrix manipulation algorithm; Basic method, use, Examples – Knapsack problem, Job sequencing with deadlines, Minimum cost spanning ree by Prim's and Kruskal's algorithm.	7
3	**Dynamic Programming and Backtracking** Basic method, use, Examples – Matrix Chain Manipulation, All pair shortest paths, single source shortest path. Basic method, use, Examples – 8 queens problem, Graph coloring problem.	7
4.	Graph traversal algorithm: Breadth First Search(BFS) and Depth First Search(DFS) – Classification of edges - tree, forward, back and cross edges – complexity and comparison, topological sorting. Network Flow: Ford Fulkerson algorithm, Max-Flow Min-Cut theorem (Statement and Illustration).	10
5.	Notion of NP-completeness: P class, NP class, NP hard class, NP complete class –	7

	their interrelationship, Satisfiability problem, Cook's theorem (Statement only), vertex cover problem and independent set problem. Approximation Algorithms: approximation scheme, polynomial time approximation schemes, vertex cover problem, travelling salesman problem.	

Text book and Reference books:

1. Introduction to Algorithms, 4TH Edition, Thomas H Cormen, Charles E Lieserson, Ronald L Rivest and Clifford Stein, MIT Press/McGraw-Hill.
2. Fundamentals of Algorithms – E. Horowitz et al.
3. Algorithm Design, 1ST Edition, Jon Kleinberg and ÉvaTardos, Pearson.
4. Algorithm Design: Foundations, Analysis, and Internet Examples, Second Edition, Michael T Goodrich and Roberto Tamassia, Wiley.
5. Algorithms -- A Creative Approach, 3RD Edition, UdiManber, Addison-Wesley, Reading, MA

Course Outcomes:	
On completion of the course students will be able to	
PCC-CSE 502.1	Analyzing the basic relation and the performance of an algorithm in terms of time complexity
PCC-CSE 502.2	Apply the algorithmic design paradigm (Divide and Conquer, Greedy method, dynamic programming and back tracking)
PCC-CSE 502.3	Design algorithm for Graph and Network Flow
PCC-CSE 502.4	Understand NP class of Problem and propose Approximation Algorithm for the same

Mapping of COs with POs and PSOs (Course Articulation Matrix):

	P01	P02	P03	PO4	PO5	PO6	PO7	PO8	PO9	P010	PO11	P012	PSO1	PSO2	PSO3
CO1	3	2	3	3	-	-	2	-	-	2	-	2	2	3	3
CO2	3	2	2	3	2	-	2	-	-	2	-	2	2	3	3
CO3	3	3	3	3	2	-	2	-	-	2	-	2	3	3	3
CO4	3	2	3	2	2	-	2	-	1	2	-	2	3	3	3
AVG.	3	2.25	2.75	2.75	2	0	2	0	1	2	0	2	2.5	3	3

Artificial Intelligence
Code: PEC-CSE 501A
Contacts: 3L
Theory: 3 hrs. / Week
Credit Points: 3

Objective:	
1.	To offer a solid basis of underlying ideas in artificial intelligence
2.	To give a brief explanation of the objectives and workings of artificial intelligence
3.	To provide the student the ability to use these methods in applications including perception, logic, and education

Pre-Requisite:	
1.	Discrete Mathematics
2.	Understanding of Probability
3.	Basic understanding of Programming Languages

Unit	Content	Hrs.
1.	**Introduction**: Fundamentals of Artificial intelligence- Issues with AI, AI technique, Tic - Tac - Toe problem. **Intelligent Agents:** Agents & environment, nature of environment, structure of agents, goal based agents, utility based agents, learning agents. **Problem Solving:** Problems, Problem Space & search: Defining the problem as state space search, production system, problem characteristics, issues in the design of search programs.	8
2.	**Search techniques:** Solving problems by searching: problem solving agents, searching for solutions; uniform search strategies: breadth first search, depth first search, depth limited search, bidirectional search, comparing uniform search strategies. **Heuristic search strategies:** Greedy best-first search, A* search, memory bounded heuristic search: local search algorithms & optimization problems: Hill climbing search, simulated annealing search, local beam search, genetic algorithms; constraint satisfaction problems, local search for constraint satisfaction problems. **Adversarial search:** Games, optimal decisions & strategies in games, the minimax search procedure, alpha-beta pruning, additional refinements, iterative deepening.	13
3.	**Knowledge & reasoning:** Knowledge representation issues, representation & mapping, approaches to knowledge representation, issues in knowledge representation.	3
4.	**Using predicate logic:** Representing simple fact in logic, representing instant & ISA relationship, computable functions & predicates, resolution, natural deduction. **Probabilistic reasoning:** Representing knowledge in an uncertain domain, the semantics of Bayesian networks, Dempster-Shafer theory, Fuzzy sets & fuzzy logics.	6
5.	**Natural Language processing:** Introduction, Syntactic processing, semantic analysis, discourse & pragmatic processing. **Learning:** Forms of learning, inductive learning, learning decision trees, explanation based learning, learning using relevance information, neural net learning & genetic learning.	6

Text book and Reference books:

1. Artificial Intelligence, Ritch & Knight, TMH
2. Artificial Intelligence A Modern Approach, Stuart Russel Peter Norvig Pearson
3. Introduction to Artificial Intelligence & Expert Systems, Patterson, PHI
4. Poole, Computational Intelligence, OUP
5. Logic & Prolog Programming, Saroj Kaushik, New Age International
6. Expert Systems, Giarranto, VIKAS
7. Stuart Russel, Peter Norvig, “AI – A Modern Approach”, Second Edition, Pearson Education, 2007.

Course Outcomes:	
On completion of the course students will be able to	
PEC-CSE 501A.1	Explain the modern tools of Artificial Intelligence (AI) and knowledge representation models.
PEC-CSE 501A.2	Determine the domain-specific problems solving methods using AI based algorithms and techniques.
PEC-CSE 501A.3	Evaluate knowledge based representation and learning methods for broad areas of state-of-the-art technological growth.
PEC-CSE 501A.4	Formulate the solution requirements of complex engineering problems in multidisciplinary application fields.

Mapping of COs with POs and PSOs (Course Articulation Matrix):

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	2	2	3	-	-	-	1	-	-	1	3	3	1
CO2	3	3	2	1	2	-	-	-	-	-	-	2	3	2	1
CO3	2	3	1	2	2	-	-	-	1	-	2	3	2	2	3
CO4	1	3	3	2	3	-	-	2	-	1	-	2	3	2	-
AVG.	2	2.75	2	1.75	2.5	0	0	2	1	1	2	2	2.75	2.25	1.67

Human Computer Interaction
Code: PEC-CSE 501B
Contact: 3L
Theory: 3 hrs. / Week
Credit Points: 3

Course Objective:	
1.	Learn the foundations of Human Computer Interaction.
2.	Be familiar with the design technologies for individuals and persons with disabilities.
3.	Be aware of mobile Human Computer interaction.
4.	Learn the guidelines for user interface.

Pre-Requisite:	
1.	The course catalog lists "knowledge of C programming language/UNIX." In reality, the prerequisite is programming skill in some practical programming language such as Java, C#, HTML, or Processing.
2.	The course focuses on human-computer interaction and interface design and assumes that students will have the skills required to program prototypes of computer interfaces.
3.	Students without programming experience who wish to take this course should speak with the instructor before the second class.

Unit	Content	Hrs.
1.	Human: I/O channels – Memory – Reasoning and problem solving; The computer: Devices – Memory – processing and networks; Interaction: Models – frameworks – Ergonomics – styles – elements – interactivity- Paradigms.	7
2.	Interactive Design basics – process – scenarios – navigation – screen design – Iteration and prototyping. HCI in software process – software life cycle – usability engineering – Prototyping in practice – design rationale. Design rules – principles, standards, guidelines, rules. Evaluation Techniques – Universal Design.	7
3.	Cognitive models –Socio-Organizational issues and stake holder requirements – Communication and collaboration models-Hypertext, Multimedia and WWW.	8

4.	Mobile Ecosystem: Platforms, Application frameworks- Types of Mobile Applications: Widgets, Applications, Games- Mobile Information Architecture, Mobile 2.0, Mobile Design: Elements of Mobile Design, Tools.	7
5.	Designing Web Interfaces – Drag & Drop, Direct Selection, Contextual Tools, Overlays, Inlays and Virtual Pages, Process Flow. Case Studies.	6
6.	Recent Trends: Speech Recognition and Translation, Multimodal System.	3

Text books:

1. Theodor Richardson, Charles N Thies, Secure Software Design, Jones & Bartlett
2. Kenneth R. van Wyk, Mark G. Graff, Dan S. Peters, Diana L. Burley, Enterprise Software Security, Addison Wesley.
3. A Dix, Janet Finlay, G D Abowd, R Beale, Human-Computer Interaction, 3rd Edition, Pearson Publishers, 2008.
4. Shneiderman, Plaisant, Cohen and Jacobs, Designing the User Interface: Strategies for Effective Human Computer Interaction, 5th Edition, Pearson Publishers, 2010.

Reference books:

1. Brian Fling, ―Mobile Design and Development, First Edition, O'Reilly Media Inc., 2009 (UNIT – IV)
2. Bill Scott and Theresa Neil, ―Designing Web Interfaces, First Edition, O'Reilly, 2009. (UNIT-V).

Course Outcomes:	
On completion of the course students will be able to	
PEC-CSE 501B.1	Understand the design and development processes and life cycle of Human Computer Interaction.
PEC-CSE 501B.2	Apply the interface design standards/guidelines for cross cultural and disabled users.
PEC-CSE 501B.3	Analyze the interface design standards/guidelines for cross cultural and disabled users.
PEC-CSE 501B.4	Design and Develop Human Computer Interaction in proper architectural structures.

Mapping of COs with POs and PSOs (Course Articulation Matrix):

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	2	1	1	3	2	2	3	2	2	2	3	2	3
CO2	3	2	1	3	2	2	-	3	3	3	3	3	3	1	-
CO3	2	2	3	2	2	1	-	1	3	3	3	1	2	-	1
CO4	3	2	2	2	-	2	3	1	3	2	3	2	-	3	1
AVG.	2.50	1.75	2.00	2.00	1.66	2.00	2.50	1.75	3.00	2.50	2.75	2.00	2.66	2.00	1.66

Computer Graphics
Code: PEC-CSE 501C
Contacts: 3L
Theory: 3 hrs. / Week
Credit Points: 3

Objective:	
1.	The main objective of the course is to introduce students with fundamental concepts and theory of computer graphics.
2.	It presents the important drawing algorithm, polygon fitting, clipping and 2D transformation curves and an introduction to 3D transformation.
3.	To make the students know about various curve and surface representation methods
4.	To make the students understand about various hidden surface removal algorithms, and lighting and shading models.

Pre-Requisite:	
1.	Knowledge of data structures and algorithm is preferable
2.	Knowledge of Linear Algebra

Unit	**Content**	**Hrs.**
1.	**Introduction to computer graphics & graphics systems**: Overview of computer graphics, representing pictures, preparing, presenting & interacting with pictures for presentations; Visualization & image processing; RGB colour model, direct coding, lookup table; storage tube graphics display, Raster scan display, 3D viewing devices, Plotters, printers, digitizers, Light pens etc.; Active & Passive graphics devices; Computer graphics software.	6
2.	**Scan conversion:** Points & lines, Line drawing algorithms; DDA algorithm, Bresenham's line algorithm, Circle generation algorithm; Ellipse generating algorithm; scan line polygon, fill algorithm, boundary fill algorithm, flood fill algorithm.	8
3.	**2D transformation & viewing**: Basic transformations: translation, rotation, scaling; Matrix representations & homogeneous coordinates, transformations between coordinate systems; reflection shear; Transformation of points, lines, parallel lines, intersecting lines. Viewing pipeline, Window to view port co-ordinate transformation, clipping operations, point clipping, line clipping, clipping circles, polygons & ellipse.	10
4.	Cohen and Sutherland line clipping, Sutherland-Hodgeman Polygon clipping, Cyrus-beck clipping method 3D transformation & viewing [5L]: 3D transformations: translation, rotation, scaling & other transformations. Rotation about an arbitrary axis in space, reflection through an arbitrary plane; general parallel projection transformation; clipping, view port clipping, 3D viewing.	6
5.	**Curves [3L]:** Curve representation, surfaces, designs, Bezier curves, B-spline curves, end conditions for periodic B-spline curves, rational B-spline curves. **Hidden surfaces [3L]:** Depth comparison, Z-buffer algorithm, Back face detection, BSP tree method, the Painter's algorithm, scan-line algorithm; Hidden line elimination, wire frame methods , fractal - geometry. **Colour & shading models [2L]:** Light & colour model; interpolative shading model; Texture. **Introduction to Ray-tracing [3L]:** Human vision and colour, Lighting, Reflection and transmission models.	6

Text book and Reference books:

1. Hearn, Baker – "Computer Graphics (C version 2nd Ed.)" – Pearson education

2. Z. Xiang, R. Plastock – "Schaum's outlines Computer Graphics (2nd Ed.)" – TMH
3. D. F. Rogers, J. A. Adams – "Mathematical Elements for Computer Graphics (2nd Ed.)" – TMH
4. Foley, J. D., Dam, A.V, Feiner, S. K., & Hughes, J. F. (1995). Computer Graphics: Principles and Practice in C. 2nd edition. Addison-Wesley Professional.
5. Bhattacharya, S. (2018). Computer Graphics. Oxford University Press
6. Marschner, S., & Shirley, P. (2017) Fundamentals of Computer Graphics. 4th edition. CRC Press

Course Outcomes:	
On completion of the course students will be able to	
PEC-CSE 501C.1	Understanding computer graphics system, display devices and various application areas of graphics.
PEC-CSE 501C.2	Applying scan conversion algorithms for line, circle and ellipse with examples.
PEC-CSE 501C.3	Analyzing and illustrating 2D and 3D transformation operations such as translation, rotation, scaling, etc.
PEC-CSE 501C.4	Evaluating any kind of 3D objects using viewing, clipping and projection techniques.

Mapping of COs with POs and PSOs (Course Articulation Matrix):															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1	-	-	-	-	-	-	-	1	-	2	-	1	-
CO2	3	2	2	3	1	-	-	1	2	-	1	2	1	2	-
CO3	3	3	3	2	2	2	1	-	2	-	-	3	2	1	2
CO4	3	3	3	3	2	-	-	2	3	-	1	2	2	2	-
AVG.	3.00	2.67	2.60	2.80	1.80	2.00	1.00	1.33	2.20	1.00	1.25	2.33	1.60	1.67	2.00

Operations Research
Code: OEC-CSE 501A
Contact: 3L
Theory: 3 hrs. / Week
Credit Points: 3

Course Objective:	
1.	To comprehend and use operations research methods to analyse and solve your operational challenges in real life.
2.	To formulate and apply the techniques of Linear Programming and the extended topics to solve certain optimization problems

Pre-Requisite:	
1.	Knowledge of probability distributions and statistics, and preferably basic calculus, for learning Simulation.

Unit	Content	Hrs.
1.	**Linear Programming Problem (LPP):** Solution of Linear Programming Problems: Solution of LPP: Using Simultaneous Equations and Graphical Method; Definitions: Feasible Solution, Basic and non-basic Variables, Basic Feasible Solution, Degenerate and Non-degenerate Solution, Convex set and explanation with examples Solution of LPP by Simplex Method; Charnes' Big-M Method; Duality Theory.	12
2.	**Transportation Problem:** Basic concept, Different solution methods, Optimality test.	9

	Assignment Problem: Problem formulation, Solution methods, Travelling Salesman Problem. **Game Theory**: Introduction; 2-Person Zero-sum Game; Saddle Point; Mini-Max and Maxi-Min Theorems (statement only) and problems; Games without Saddle Point; Graphical Method; Principle of Dominance.	
3.	**Network Analysis**: Shortest Path: Floyd Algorithm; Maximal Flow Problem: Ford-Fulkerson; PERT & CPM (Cost Analysis, Crashing, Resource Allocation excluded).	6
4.	**Queuing Theory**: Introduction; Basic Definitions and Notations; Axiomatic Derivation of the Arrival & Departure (Poisson Queue). Poisson Queue Models: (M/M/1): (∞ / FIFO) and (M/M/1: N / FIFO) and problems. **Inventory Control**: Introduction to EOQ Models of Deterministic and Probabilistic; Safety Stock; Buffer Stock. **Sequencing:** Johnson's algorithm	9

Text book and Reference books:

1. H. A. Taha, "Operations Research", Pearson
2. P. M. Karak – "Linear Programming and Theory of Games", ABS Publishing House
3. Ghosh and Chakraborty, "Linear Programming and Theory of Games", Central Book Agency
4. Ravindran, Philips and Solberg - "Operations Research", WILEY INDIA

Course Outcome:	
On completion of the course students will be able to	
OEC-CSE 501A.1	Recall the fundamental principles and formulas related to linear algebra and linear inequalities.
OEC-CSE 501A.2	Demonstrate a comprehensive understanding of the theoretical principles and applications of linear algebra, convex & concave combination, graph theory and stochastic process necessary for engineering practice by enhancing the power of knowledge.
OEC-CSE 501A.3	Apply advanced techniques in LPP, Transportation problem, Game theory, Network analysis, Inventory theory and Queuing theory to solve complex mathematical problems and analyze linear dependency & independency, types of solutions, loop formation, steady state in optimization and multivariable analysis.
OEC-CSE 501A.4	Analyze multidisciplinary problems using different mathematical models and create a model and build a path by which a complex multidisciplinary engineering problem can be solved.

Mapping of COs with POs and PSOs (Course Articulation Matrix):

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	3	2	-	-	-	-	-	2	2	3	3	2
CO2	3	3	3	3	2	-	-	-	-	-	2	3	3	3	3
CO3	3	3	3	3	3	-	-	-	-	-	2	3	3	3	3
CO4	3	3	2	3	3	-	-	-	-	-	2	2	3	3	2
AVG.	3.00	3.00	2.50	3.00	2.50	-	-	-	-	-	2.00	2.50	3.00	3.00	2.50

Multimedia Systems
Code: OEC-CSE 501B
Contacts: 3L
Theory: 3 hrs. / Week

Course Objective:	
1.	To understand various components of the multimedia systems
2.	To introduce how multimedia can be used in various application areas.
3.	To provide a solid foundation to the students so that they can identify the proper applications of multimedia, evaluate the appropriate multimedia systems and develop effective multimedia applications.

Pre-Requisite:	
1.	Knowledge in Data Structure, Computer Network and Operating System.

Unit	Content	Hrs.
1.	**Introduction**: Multimedia today, Impact of Multimedia, Multimedia Systems, Components and Its Applications.	2
2.	**Text and Audio, Image and Video:** Text: Types of Text, Ways to Present Text, Aspects of Text Design, Character, Character Set, Codes, Unicode, Encryption; Audio: Basic Sound Concepts, Types of Sound, Digitizing Sound, Computer Representation of Sound (Sampling Rate, Sampling Size, Quantization), Audio Formats, Audio tools, MIDI Image: Formats, Image Color Scheme, Image Enhancement; Video: Analogue and Digital Video, Recording Formats and Standards (JPEG, MPEG, H.261) Transmission of Video Signals, Video Capture, and Computer based Animation.	10
3.	**Synchronization, Storage models and Access Techniques**: Temporal relationships, synchronization accuracy specification factors, quality of service, Magnetic media, optical media, file systems (traditional, multimedia) Multimedia devices – Output devices, CD-ROM, DVD, Scanner, and CCD.	8
4.	**Image and Video Database, Document Architecture and Content Management:** Image representation, segmentation, similarity based retrieval, image retrieval by color, shape and texture; indexing- kd trees, R-trees, quad trees; Case studies- QBIC, Virage. Video Content, querying, video segmentation, indexing, Content Design and Development, General Design Principles Hypertext: Concept, Open Document Architecture (ODA), Multimedia and Hypermedia Coding Expert Group (MHEG), Standard Generalized Markup Language (SGML), Document Type Definition (DTD), Hypertext Markup Language (HTML) in Web Publishing. Case study of Applications.	12
5.	**Multimedia Applications**: Interactive television, Video-on-demand, Video Conferencing, Educational Applications, Industrial Applications, Multimedia archives and digital libraries, media editors.	5

Text book and Reference books:

1. Ralf Steinmetz and Klara Nahrstedt, Multimedia: Computing, Communications & Applications, Pearson Ed.
2. Nalin K. Sharda, Multimedia Information System, PHI.
3. Fred Halsall, Multimedia Communications, Pearson Ed.
4. Koegel Buford, Multimedia Systems, Pearson Ed.
5. Fred Hoffstetter, Multimedia Literacy, McGraw Hill.
6. Ralf Steinmetz and Klara Nahrstedt, Multimedia Fundamentals: Vol. 1- Media Coding and Content Processing, PHI.
7. J. Jeffcoate, Multimedia in Practice: Technology and Application, PHI.

8. V.K. Jain, Multimedia and Animation, Khanna Publishing House, New Delhi (AICTE Recommended Textbook – 2018)

Course Outcome:	
On completion of the course students will be able to	
OEC-CSE 501B.1	Analysing the range of concepts, techniques and tools for creating and editing the interactive multimedia applications
OEC-CSE 501B.2	To understand the hardware and software needed to create projects using creativity and organization to create them.
OEC-CSE 501B.3	To apply the concepts of Synchronization, Storage models and Access Techniques.
OEC-CSE 501B.4	To evaluate the approaches for Image and Video Database, Document Architecture , Content Management to develop a complete project.

Mapping of COs with POs and PSOs (Course Articulation Matrix):

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	3	3	1	1	1	1	1	1	2	2	1	2
CO2	2	2	3	2	2	2	1	-	2	1	2	3	-	2	1
CO3	2	1	3	3	2	2	-	-	2	-	2	3	2	3	2
CO4	3	2	2	3	2	-	1	2	1	2	3	2	3	1	2
AVG.	2.50	1.75	2.50	2.75	2.25	1.25	0.75	0.75	1.50	1.00	2.00	2.50	2.33	2.17	1.40

Introduction to Philosophical Thoughts
Code: OEC-CSE 501C
Contact: 3L
Theory: 3 hrs. / Week
Credit Points: 3

Course Objective:	
1.	To inspire the student to confront the philosophical problems implicit in the experience of self, others and the universe, together with the question of their relations to ultimate transcendence (God and immortality);
2.	To develop in the student habits of clear, critical thinking within the framework of both an adequate philosophical methodology and accepted norms of scholarship;
3.	To introduce the student to reading critically the great philosophers, past and present
4.	Finally, to help the student to formulate for himself or herself a philosophy of life or world-view consistent with the objectives of liberal education at a Catholic university.

Pre-Requisite:	
1.	Multiple Measures Placement in GE-level writing, or completion of 113A or 114A, or completion of the lower division writing requirement.

Unit	Content	Hrs.
1.	**Nature of Indian Philosophy**: Plurality as well as common concerns. 2. Basic concepts of the Vedic and Upanisadic views: Atman, Jagrata, Svapna, Susupti, Turiya, Brahman, Karma, Rta, Rna.	10
2.	**Carvaka school**: its epistemology, metaphysics and ethics. Mukti	9
3.	**Jainism**: Concepts of sat, dravya, guna, paryaya, jiva, ajiva, anekantavada, syadvada, and nayavada; pramanas, ahimsa, bondage and liberation.	7
4.	Buddhism: theory of pramanas, theory of dependent origination, the four noble truths; doctrine of momentaryness; theory of no soul. The interpretation of these	5

	theories in schools of Buddhism: Vaibhasika, Sautrantrika, Yogacara, Madhyamika.	
5.	Nyaya: theory of Pramanas; the individual self and its liberation; the idea of God and proofs for His existence.	5

Text book and Reference books:

1. M. Hiriyanna: Outlines of Indian Philosophy.
2. C. D. Sharma: A Critical Survey of Indian Philosophy.
3. S. N. Das Gupta: A History of Indian Philosophy Vol – I to V.
4. S. Radhakrishnan: Indian Philosophy Vol – I & II.
5. T. R. V. Murti: Central Philosophy of Buddhism.
6. J. N. Mahanty: Reason and Tradition of Indian Thought.
7. R. D. Ranade: A Constructive Survey of Upanisadic Philosophy.
8. P. T. Raju: Structural Depths of Indian Thought.
9. K. C. Bhattacharya: Studies in Philosophy Vol – 1.
10. Datta and Chatterjee: Introduction of Indian Philosophy

Course Outcome:	
On completion of the course students will be able to	
OEC-CSE 501C.1	Understand and explain philosophically important theories and concepts that have historically been used to organize and explain human experience.
OEC-CSE 501C.2	Apply philosophical methods and principles to analyse real-world issues and dilemmas.
OEC-CSE 501C.3	Analyse the arguments presented in philosophical texts and contemporary philosophical discourse.
OEC-CSE 501C.4	Evaluate the strengths and weaknesses of various philosophical theories and positions.

Mapping of COs with POs and PSOs (Course Articulation Matrix):

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	3	1	1	2	2	3	1	2	-	3	2	2	2
CO2	-	-	2	2	-	1	2	3	2	1	-	3	2	1	2
CO3	-	-	2	1	-	2	2	3	1	2	-	3	1	2	2
CO4	-	-	2	-	-	2	2	3	1	2	-	3	2	2	2
AVG.	1.00	1.00	2.25	1.33	1.00	1.75	2.00	3.00	1.25	1.67	1.00	3.00	1.75	1.75	2.00

Project Management and Entrepreneurship (Humanities-IV)
Code: HSMC-CSE 501
Contact: 2L+1T
Theory: 3 hrs. / Week
Credit Points: 3

Course Objective:	
1.	The course aims to bridge the gap between project management principles and entrepreneurial practices, enabling students to apply project management methodologies within the dynamic and innovative context of entrepreneurship.
2.	Another key objective is to foster the development of entrepreneurial skills, such as creativity, leadership, and adaptability, and cultivate an entrepreneurial mind-set characterized by resilience, risk-taking, and opportunity recognition.
3.	The course seeks to prepare students for launching and managing start-up ventures by equipping

	them with the knowledge, tools, and practical experience needed to identify viable business opportunities, develop comprehensive business plans, and execute entrepreneurial projects effectively.

Unit	Content	Hrs.
1.	**Basic Concepts of Project Management:** Definition and types of project, Project life cycle, Project constraints, Organizational structures for projects, Responsibilities of project manager, Project risk analysis, Project appraisal.	4
2.	**Environmental and social aspects of project:** Environmental considerations in project evaluation, Primary issues and secondary issues in Feasibility study, Social cost benefit analysis, Managing Project Risks, Contingency Planning, Project Audit Process and Closure.	4
3.	**Network analysis:** Network modeling of a project, Activity on Arrow (AOA), Forward and backward pass computation, Critical paths, floats and slack, Project Scheduling Techniques, PERT, CPM Models, Time-Cost Trade-off in a project, Project Monitoring Techniques, Gnatt chart, Line of Balance (LoB).	10
4.	**Foundation of Entrepreneurship:** Entrepreneurship and Intrapreneurship (comparisons), India's Start up Revolution, Rural and Social Entrepreneurship, Key attributes of an entrepreneur, women entrepreneurship, Myths and realities of entrepreneurship, Transition from college/ regular job to the world startups.	6
5.	**Project Financing for entrepreneurs:** Venture Capitalist, Role of Incubators and Angel Investors, Private Equity, Crowd sourcing, Overview of debt financing, Examples of project management softwares.	6

Text Books and References

1. Project Management by Prasanna and Chandra, Tata McGraw Hill.
2. Elements of Project Management by Pete Spinner, Prentice Hall, USA.
3. A course in PERT and CPM by R. C. Gupta, Dhanpat Rai Publications(P) Ltd, Delhi.
4. Project Management for Engineering, Business and Technology: Nicholas, J.M., and Steyn, H.; PHI
5. Innovation and Entrepreneurship by Drucker, P.F.; Harper and Row
6. Business, Entrepreneurship and Management: Rao, V. S. P Vikas

Course Outcome:	
On completion of the course students will be able to	
HSMC-CSE 501.1	Students will demonstrate the ability to integrate project management principles, methodologies, and tools with entrepreneurial practices to effectively plan, execute, and manage projects within startup ventures.
HSMC-CSE 501.2	Students will be able to identify and evaluate business opportunities by applying project management techniques such as feasibility analysis, market research, and risk assessment, enabling them to make informed decisions regarding new venture creation.
HSMC-CSE 501.3	Students will develop an entrepreneurial mindset characterized by creativity, innovation, and adaptability, as well as leadership skills necessary to drive entrepreneurial initiatives, lead project teams and navigate uncertainties inherent in startup environments.
HSMC-CSE 501.4	Students will acquire the knowledge and skills needed to effectively execute and manage entrepreneurial projects, including project planning, resource allocation, time management and stakeholder communication, ensuring successful project delivery within budget and schedule constraints.

Mapping of COs with POs and PSOs (Course Articulation Matrix):															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
3	3	1	3	2	1	1	1	1	1	3	2	3	3	1	3
3	3	2	2	2	1	1	1	1	1	3	2	3	3	2	3
2	2	3	3	3	2	2	2	3	2	2	2	3	3	1	2
1	2	3	3	3	2	2	3	3	3	2	2	3	3	2	1
2.25	2.5	2.25	2.75	2.5	1.5	1.5	1.75	2	1.75	2.5	2	3	3	1.5	2.25

Operating System Lab
Code: PCC-CSE 591
Contacts: 4P
Practical: 4 hrs. / Week

Course Objective:	
1.	To install windows operating systems.
2.	To understand the basics of Unix command and shell programming.
3.	To implement various CPU scheduling algorithms.
4.	To implement Deadlock Avoidance and Deadlock Detection Algorithms
5.	To implement Page Replacement Algorithms
6.	To implement various memory allocation methods.

Pre-Requisite:	
1.	Basic programming knowledge.

Unit	Content	No. of sessions
1.	**Managing Unix/Linux Operating System:** Creating a bash shell script, making a script executable, shell syntax (variables, conditions, control structures, functions, and commands). Partitions, Swap space, Device files, Raw and Block files, Formatting disks, Making file systems, Superblock, I-nodes, File system checker, Mounting file systems, Logical Volumes, Password file management, Password security, Groups and the group file, Shells, restricted shells, user-management commands, homes and permissions, default files, profiles, locking accounts, setting passwords, Switching user, Switching group, Removing users & user groups.	5
2.	**Process:** Starting new process, implement child, parent and grandparent process, waiting for a process, zombie process, orphan process.	2
3.	**Signal:** Signal handling, sending signals, signal interface, signal sets.	2
4.	**CPU scheduling algorithms:** Implement various cpu scheduling algorithm using C programming language.	2
5.	**Threads:** programming with pthread functions (viz. pthread_create, pthread_join, pthread_exit, pthread_attr_init, pthread_cancel)	1
6.	**Inter-process communication:** Pipes (use functions pipe, popen, pclose), named pipes (FIFOs, accessing FIFO), message passing & shared memory (IPC).	1

(Detailed instructions for Laboratory Manual to be followed for further guidance)

Text book and Reference books:

1. Richard Stevens, Stephen Rago, Advanced Programming in the UNIX Environment, Pearson Education, 2/e.
2. UNIX : Concepts and Applications | 4th Edition by Sumitabha Das.
3. The C Odyssey: UNIX: v. 3 by Vijay Mukhi.

Course Outcomes:	
On completion of the course students will be able to	
PCC-CSE 591.1	Understand elementary UNIX system commands used for UNIX operating system.
PCC-CSE 591.2	Apply the knowledge of shell program to solve various complex problems.
PCC-CSE 591.3	Evaluate process creation concepts, different CPU scheduling algorithms and user threads.
PCC-CSE 591.4	Design different IPC techniques and synchronization problems.

Mapping of COs with POs and PSOs (Course Articulation Matrix):															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	2	1	2	1	1	1	2	1	1	1	2	2	2	2
CO2	2	2	2	2	2	-	2	-	2	2	2	2	3	2	3
CO3	2	2	2	2	2	-	2	-	2	2	2	2	2	2	3
CO4	2	2	3	3	2	-	2	-	2	2	2	2	2	2	3
AVG.	1.75	2.00	2.00	2.25	1.75	0.25	1.75	0.50	1.75	1.75	1.75	2.00	2.25	2.00	2.75

Design and Analysis of Algorithm Lab
Code: PCC-CSE 592
Contacts: 4P
Practical: 4 hrs. / Week
Credit Points: 2

Course Objective:	
1.	Ability to develop C programs for computing and real-life applications using basic elements like control statements, arrays, functions, pointers and strings, and data structures like stacks, queues and linked lists.
2.	Ability to Implement searching and sorting algorithms.

Pre-Requisite:	
1.	Pre-requisites as in PCC-CS301

Laboratory Experiments:	
Divide and Conquer:	
1.	Implement Binary Search using Divide and Conquer approach Implement Merge Sort using Divide and Conquer approach.
2.	Implement Quick Sort using Divide and Conquer approach Find Maximum and Minimum element from an array of integer using Divide and Conquer approach.
Dynamic Programming:	
3.	Find the minimum number of scalar multiplication needed for chain of matrix
4.	Implement all pair of Shortest path for a graph (Floyed- Warshall Algorithm) Implement Traveling Salesman Problem
5.	Implement Single Source shortest Path for a graph (Dijkstra , Bellman Ford Algorithm
Backtracking:	
6.	Implement 8 Queen problem
7.	Graph Coloring Problem

Greedy method:	
8.	Knapsack Problem
9.	Minimum Cost Spanning Tree by Prim's Algorithm Minimum Cost Spanning Tree by Kruskal's Algorithm
Graph Traversal Algorithm:	
10.	Implement Breadth First Search (BFS) Implement Depth First Search (DFS)

(Detailed instructions for Laboratory Manual to be followed for further guidance)

Course Outcomes:	
On completion of the course students will be able to	
PCC-CSE 592.1	**Analyze** different types of applications of Divide & Conquer techniques
PCC-CSE 592.2	**Understanding** the implementation of Dynamic Programming techniques
PCC-CSE 592.3	**Apply** backtracking and Greedy method for different types of algorithm
PCC-CSE 592.4	**Evaluate** and Implement the Graph Traversal Algorithms

Mapping of COs with POs and PSOs (Course Articulation Matrix):

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	3	3	-	-	2	-	-	2	-	2	2	3	3
CO2	3	2	2	3	2	-	2	-	-	2	-	2	2	3	3
CO3	3	3	3	3	2	-	2	-	-	2	-	2	3	3	3
CO4	3	2	3	2	2	-	2	-	1	2	-	2	3	3	3
AVG.	3	2.25	2.75	2.75	2	0	2	0	1	2	0	2	2.5	3	3

IT Workshop-II (Web Programming)
Code: PCC-CSE 593
Contacts: 4P
Practical: 4 hrs. / Week
Credit Points: 2

Course Objective:	
1.	To introduce scripting language concepts for developing client-side applications.
2.	Comprehend the basics of the internet and web terminologies.
3.	Familiar with client server architecture and able to develop web applications.
4.	Students will gain the skills to develop project-based experience .

Pre-Requisite:	
1.	Basic Object Oriented Programming Concepts are required.
2.	How the internet works, including concepts like IP addresses, DNS, and how web browsers interact with servers.
3.	Knowing how HTTP requests and responses work.

Unit	Content	Hrs.
1.	DHTML (HTML , CSS , Java Script): HTML: Introduction, Editors, Elements, Attributes, Heading, Paragraph. Formatting, Link, Head, Table, List, Block, LayoutForm, Iframe, Colors, Colorname, Colorvalue.Image Maps: map, area, attributes of image area. CSS: Panels , Borders , Padding , Margins , Display,Buttons ,Notes , Quotes , Alerts , Images , Tags , Icons. Java Script : Statements, Syntax, Comments , Operators, Data Types , Functions , Objects , Events, Strings , Arrays , Math, Errors , Scope , Classes , Modules.	4
2.	AngularJS: Directives, Model, Data Binding , Controllers, Scopes , Filters , DOM , Events, Forms, Validation, API.	2
3.	Node.js: Modules, HTTP Module, File System , URL Module, NPM ,Events , Upload Files , Email.	2
4.	Applet: Preparing to write Applets, Building Applet code, Applet Tag, Passing parameters to Applets, Adding Applet to HTML file.	2
5.	PHP&MySql: PHP :Install ,Syntax , Comments, Variables , Echo / Print, Data Types, Strings, Numbers, Casting, Math, Constants, Magic Constants, Operators ,If...Else...Elseif, Switch, Loops, Functions , Arrays , Superglobals , RegEx ,Forms ,Form Handling , Form Validation. MySql :Database, Connect, Create DB, Create Table, Insert Data, Get Last ID, Insert Multiple, Prepared, Select Data, Order By, Delete Data, Update Data, Limit Data.	3
6.	Extensible Markup Language: Introduction, Tree, Syntax, Elements, Attributes, Validation, Viewing. XHTML in brief. CGI Scripts: Introduction, Environment Variable, GET and POST Methods.	1

Text book and Reference books:

1. Web Technology: A Developer's Perspective, N.P. Gopalan and J. Akilandeswari, PHI Learning, Delhi, 2013. (Chapters 1-5, 7, 8, 9).
2. Internetworking Technologies, An Engineering Perspective, Rahul Banerjee, PHI Learning, Delhi, 2011.
3. AchyutGodbole,AtulKahate"WebTechnologies:TCP/IP,Web/JavaProgramming, and CloudComputing",Third Edition,McGrawHill Education.

Course Outcome:	
On completion of the course students will be able to	
PCC-CSE 593.1	Understand and develop a dynamic webpage by the use of java script and DHTML
PCC-CSE 593.2	Apply and evaluate the need for secured web application development with client-side, server-side scripting languages.
PCC-CSE 593.3	Analyse web programming and connection with DBMS and perform different SQL operations using MySql.
PCC-CSE 593.4	To create small interactive web modules using modern tools following the professional web based engineering solutions, ethics and management techniques.

Mapping of COs with POs and PSOs (Course Articulation Matrix):

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	2	2	2	1	1	1	-	1	-	2	2	2	1
CO2	2	2	3	2	1	-	-	2	2	-	2	-	1	1	1
CO3	1	3	2	3	2	2	2	-	3	-	1	1	2	3	2
CO4	1	1	2	2	2	1	-	2	-	2	3	2	1	1	1
AVG.	1.50	2.00	2.25	2.25	1.75	1.00	0.75	1.00	1.25	0.75	1.50	1.25	1.50	1.75	1.25